

Rituraj Singh

AI Engineer • LLM Systems, Agentic AI & Python Backends

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SUMMARY

Final-year B.Tech (Computer Science — AI & ML) student at VIT Bhopal University (Class of 2027, CGPA 8.39/10) who builds LLM systems end-to-end: retrieval over unstructured documents, tool use, schema-gated outputs, and guardrails that catch hallucinated model output before it reaches users. Ships those systems as Python API services against PostgreSQL, with an emphasis on evaluation, auditability, and safe behaviour in compliance-heavy domains.

TECHNICAL SKILLS

Generative AI & LLMs	Claude, Gemini, Llama 3.3, Groq • prompt engineering, RAG, tool use, structured outputs, guardrails, LLM evaluation & benchmarking
Agentic Systems	Multi-stage agent pipelines, tool integration, orchestration, Pydantic, Zod, async & queue-driven processing
Languages	Python, SQL, TypeScript, Java, C++
Backend & APIs	Django, DRF, FastAPI, REST, Docker, microservices, event-driven patterns
Data & Storage	PostgreSQL, multi-tenant schemas, row-level security, audit trails, OCR, time-series
ML & Vision	PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, CNNs
Cloud & Practice	AWS, Git, GitHub Actions, CI/CD, PyTest, Linux, Agile

WORK EXPERIENCE

Software Developer Intern — ConsultiFI Data (Remote)

Jun 2026 – Present

- Ship features into a production Django service — REST endpoints (DRF), ORM models and migrations for device and trip data, and scheduled jobs — inside an established codebase alongside the platform team.
- Build Python ingestion pipelines for IoT oil flow-meter telemetry, with validation that flags abnormal flow, sensor drift, and missing-data windows before the numbers reach reporting.
- Aggregate fleet telemetry (GPS, fuel draw, run-hours) into consumption, utilisation, and route-level metrics surfaced through operations dashboards.

AI Intern — Tata Steel Utilities & Infrastructure Services Ltd, Jamshedpur (On-site)

May – Jun 2025

- Deployed a computer-vision system (Python, OpenCV) across 4 live CCTV feeds at an active industrial facility, with automated real-time alerting.
- Owned frame extraction, data preparation, detection tuning, and evaluation — 92% accuracy on 1,000+ vehicle detections/day in production, cutting manual monitoring workload by ~60%.

PROJECTS

Praxya — LLM Document Intelligence for Regulated GHG Reporting • [GitHub](#)

Mar 2026 – Present

- Sole owner of the system in production** — architecture, LLM integration, database design, deployment, and ongoing monitoring.
- Designed a 9-stage pipeline — PDF → OCR → LLM extraction via tool use → Pydantic validation → calculation → auditor-ready report — turning unstructured filings into validated structured data; benchmarked providers on cost, latency, and accuracy, then moved extraction to Llama 3.3 70B on Groq.
- Implemented SEBI BRSR / IPCC Scope 1 & 2 calculations on a multi-tenant PostgreSQL schema with insert-only audit trail and row-level security, giving traceability from every reported figure back to its source document.

Conversion Funnel Analyzer — Agentic Pipeline with Guardrails • [GitHub](#)

May 2026

- Three-stage agent pipeline (map → analyze → recommend) returning schema-validated, prioritized recommendations, behind a cross-validation layer that strips hallucinated output and enforces domain invariants before users see results.

FileMyCa — GST Data-Collection SaaS for CA Firms • [GitHub](#)

Feb – Mar 2026

- Solo-shipped a B2B portal (Next.js, TypeScript, Supabase) in 5 days: admin dashboard tracking 47 MSME clients, strict upload validation, automated bulk reminders.

EDUCATION

B.Tech — Computer Science & Engineering (AI & ML), VIT Bhopal University

2023 – 2027

CGPA: 8.39 / 10 • No active backlogs • Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Cloud Computing

Class XII (CBSE) 86.4% • Class X (CBSE) 94.6% — DAV Public School, Bistupur

2021 – 2023